In-class activity 5 part 1

1. **Example of models with different number of predictors**

n = 100, $\mathbf{M_3}$: $Y \sim X_1 + X_2 + X_3$ p = 4 (intercept included)

$\mathbf{M_4}$: $Y \sim X_1 + X_2 + X_3 + X_4$ p = 5 (intercept included)

	RSS	$\boldsymbol{n * log \frac{RSS}{n}}$	**2*p**	**AIC**	**Better model**
M_3	200	$100 * log \frac{200}{100} =$ 69.31	2*4 = 8	69.31+8 = 77.31	--
M_4(A)	198	$100 * log \frac{198}{100} =$ 68.31	2*5 = 10	68.31+10 = 78.31	
M_4(B)	195	$100 * log \frac{195}{100} =$ 66.78	2*5 = 10	66.78+10 = 76.78	

M3 vs M4 (A)

1. How much does the RSS decrease from M_3 to M_4 (A)? _____
2. How much does the $\boldsymbol{n * log \frac{RSS}{n}}$ decrease from M_3 to M_4 (A)? _____
3. How much does the Penalty increase from M_3 to M_4 (A)? _____
4. Is the reduction in RSS large enough to justify adding X4? _____

M3 vs M4 (B)

5. How much does the RSS decrease from M_3 to M_4 (B)? _____
6. How much does the $\boldsymbol{n * log \frac{RSS}{n}}$ decrease from M_3 to M_4 (B)? _____
7. How much does the Penalty increase from M_3 to M_4 (B)? _____
8. Is the reduction in RSS large enough to justify adding X4? _____

2. Example of models with same number of predictors

Model A, B, C have same number of predictors p = 6.

M_A: $Y \sim X_1 + X_2 + X_3 + X_4 + X_5$
M_B: $Y \sim X_6 + X_7 + X_8 + X_9 + X_{10}$
M_C: $Y \sim X_1 + X_2 + X_7 + X_9 + X_{10}$

Answer following question using the top table (AIC table)

1) Is the penalty term (2*p) the same for all models under criteria AIC? _____
2) When comparing models A,B,C with the same number of predictors, does the selection criterion **AIC** ultimately depend on the **RSS** or on the **penalty term**? _____

Answer following question using the bottom table (BIC table)

3) Is the penalty term (p*log (n)) the same for all models under criteria BIC? ____
4) Is the penalty term (p*log (n)) under criteria BIC **heavier or lighter** than that of AIC? __
5) When comparing models A,B,C with the same number of predictors, does the selection criterion **BIC** ultimately depend on the **RSS or on the penalty term**? ___
6) When comparing models A, B, and C that have the **same number of predictors**, can different selection criteria lead to different chosen models? _____

	RSS	$n * log \frac{RSS}{n}$	2*p	AIC	Better model
M_A	200	$100 * log \frac{200}{100} = 69.31$	2*6= 12	69.31+12 = 81.31	M_C
M_B	198	$100 * log \frac{198}{100} = 68.31$	2*6 = 12	68.31+12 = 79.31	
M_C	195	$100 * log \frac{195}{100} = 66.78$	2*6 = 12	66.78+12 = 78.78	

	RSS	$n * log \frac{RSS}{n}$	p*log(n)	BIC	Better model
M_A	200	$100 * log \frac{200}{100} = 69.31$	6*log(100) = 27.63	69.31+27.63 = 96.94	M_C
M_B	198	$100 * log \frac{198}{100} = 68.31$	6*log(100) = 27.63	68.31+27.63 = 95.94	
M_C	195	$100 * log \frac{195}{100} = 66.78$	6*log(100) = 27.63	66.78+27.63 = 94.41	